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Correspondence to:
Bijaya Bikram Samal
bijaybikram@iitkgp.ac.in

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Four-dimensional (4D) printing through FDM: Effect of infill density and bed temperature on shape memory properties in different thermo-mechanical programming conditions

Bijaya Bikram Samal[1], Shailendra Kumar Varshney[2] and Cheruvu Siva Kumar[3]

[1]Advanced Technology Development Centre, Indian Institute of Technology, Kharagpur 721302, West Bengal, India, [2]Department of Electronics and Electrical Communications Engineering, Indian Institute of Technology, Kharagpur 721302, West Bengal, India, [3]Department of Mechanical Engineering, Indian Institute of Technology, Kharagpur 721302, West Bengal, India

Abstract This study investigates the impact of key parameters in fused deposition modeling (FDM), i.e., infill density and bed temperature, on the shape memory properties of four-dimensional (4D) printed shape memory polymer (SMP). Two thermo-mechanical programming methodologies, namely programming during printing (PDP) and programming after printing (PAP), are used to examine their effects too. Increasing the infill percentage greatly improves the ability of the material to regain its original shape. This is achieved by increasing the amount of material and minimizing gaps between layers, which helps to build up the pre-strain, which causes shape morphing. On the other hand, higher bed temperatures decrease the shape recovery by slowing down cooling, decreasing residual strain, and minimizing thermal stresses. Compared to PAP samples, PDP samples show a significant 40 % increase in shape memory index (SMI), indicating the considerable influence of FDM printing rather than post-printing programming process.

1. Introduction

Smart materials have been used in all scales for various applications like shape memory alloy (SMA) thin film-based thermal actuators [1], shape memory polymer (SMP) based vascular stents [2], shape memory composites for morphing wingtips, etc. [3]. Among various means to fabricate smart materials, additive manufacturing has been widely utilized to create very complex stimuli-responsive structures, paving the way for the advancement of 4D printing technology [4].

"4D printing can be defined as a method of fabricating a part layer by layer using smart materials to create a structure that can change its characteristic properties over time when an external stimulus is provided to it."

It can be briefly classified into three different types. The first one is 4D printing, which embeds active smart material [5]. The second type is 4D printing of a single smart material, its blend or composite [6]. The third type is 4D printing with multiple smart materials, its blends or composites [7, 8]. To achieve 4D printing, different additive manufacturing (AM) technologies can be utilized, but fused filament fabrication (FFF) has been more prevalent due to the ease of handling the system, economical, scalable, and broad range of smart materials available. FFF is also known as fused deposition modeling (FDM) or extrusion additive manufacturing (Ex-AM) [9]. The smart materials that can be processed using FFF/FDM are shape memory polymers (SMP). SMPs are widely used in 4D printing due to their advantages over other smart materials. They can get deformed up to 800 % with very low deformation stress, are inexpensive, easy to

process, highly tunable, etc. [10]. 4D printing was first coined by Tibbits et al. in 2013 [11]. The study utilized hydrophilic polymer for expansion up to 150 % and rigid polymer as an angle limiter in a predefined geometry, allowing shape change [12]. Bodaghi et al. [13] explored functionally graded 4D printing (FG4DP) using FDM and SMP to create adaptive metamaterials with performance-driven functionality, enabling self-folding and self-coiling. Samal et al. studied the effect of printing process parameters and revealed that higher printing speed enhances shape recovery, while increased liquefier temperature reduces it [14]. In another study, Noroozi et al. [15] explored the use of 4D printing technology in creating adaptive meta-structures, examining the impact of printing speed on shape recovery and self-bending behavior. Wu et al. [16] examined the effect of four process parameters on the shape recovery ratio (R_r) and recovery rate of polylactic acid (PLA) SMP printed using FDM, using a design of experiments (DoE) approach. Factors ranked in importance for R_r included recovery temperature (T_r), deformation temperature (T_d), raster angle, and layer thickness. The study by Hosseinzadeh et al. [17] found that temperature factors significantly improved the shape memory characteristics of PLA SMP structures, with the highest recovery achieved at a T_d of 53.31 °C, T_r of 59.94 °C, and heating rate (H_r) of 8.05 °C per minute. Researchers [18] studied and explored heat-induced shape changes in 3D-printed thermoplastics like PLA, HIPS, and ABS. It reveals that different parameters, such as printing speed, curvature radius, print orientations, and infill density, influence these transformations, providing insights for optimizing shape transformations. Barletta et al. [19] utilized PLA as stress-absorbing components in 4D printing via FDM. Results show exceptional shape memory with a recovery time (R_t) of 10 seconds and up to 99.3 % R_r at a T_r of 85 °C. However, lower nozzle temperature and layer height affect R_t. Wu et al. [20] used FDM to analyze the impact of printing speed and component geometries on self-deformation in 4D-printed shape memory PLA. It is found that higher speed and thinner samples caused greater self-deformation. The study also found a positive correlation between stress variations and bending angles [21]. Alshebly and Nafea [22], in recent research, found a positive correlation between printing speed and pre-strain whereas, an inverse relationship between higher nozzle temperatures and pre-strain. They also suggested the importance of maintaining a 50 % passive-to-active layer ratio with controlling the printed layer and actuator heights to get an enhanced shape memory property.

Tezerjani et al. [23] observed in their study that bed temperature doesn't significantly impact the 4D printing process. Recently, it has been reported [24] that high temperature and high infill density provide better recovery, whereas low temperature and low infill density result in poor recovery.

Despite substantial studies on the effect of process factors in FDM 4D printed SMP, many crucial parameters, particularly infill density and print bed temperature, remain largely unexplored and require additional independent investigation. The existing literature focuses mostly on programming during the printing (PDP) method, while the post-printing programming scenario (PAP) has received insufficient attention. As a result, the purpose of this research is to find the effect of infill density and print bed temperature on the shape memory properties of PLA SMP while taking into account two distinct programming methodologies.

2. Materials and method

2.1 Material consideration

The material used for the experimentation is commonly called polylactide or polylactic acid (PLA). PLA based SMP filament with 1.75 mm thickness is procured from eSUN® (eSUN, Shenzhen, China). Detail material property is provided by the manufacturer [25]. The glass transition temperature (T_g) of the material is provided as 60 °C. PLA's shape memory effect is due to its semi-crystalline structure, which stores elastic energy in unordered polymeric chains. When deformed, these chains recoil, releasing strain and restoring their original shape. PLA's biodegradability, biocompatibility, and ease of processing make it the second most used material for 3D printing globally.

2.2 FDM 4D printing, programming conditions, and characterization

The experimental part is selected to be an origami cuboid with a proper notch in the middle. By perceiving the geometry of the part after actuation, i.e., folding from the middle at 90 degrees, the design is considered to have a chamfer of 45 degrees in both the ends of the notch and a depth of 1 mm. The part is designed using SolidWorks software.

Once the design is ready, the 3D model is converted to an STL file for further preprocessing. The preprocessing involves the slicing of the 3D model. Slicing is done using Simplify3D software, and the G-Code file is generated for printing. Two different process parameters were varied during the slicing, the first being bed temperature and the second being infill percentage. Different values of bed temperatures and infill density considered are provided in Table 2. Apart from these two process parameters, other process parameters were kept constant, as shown in Table 1. The FFF/FDM 3D printer considered for the experiment is an Ender 3 (Shenzhen Creality 3D Technology Co., Ltd.). It is manually leveled and well-calibrated before the actual printing. Once the parts are printed, they are left to cool before being removed and taken for shape memory characterization (Fig. 1).

Infill parameters are selected arbitrarily as they can vary from 1 % to 100 %. For bed temperature, the first value is approximately the ambient temperature, and then it is varied arbitrarily till the highest possible value (100 °C) as per the machine specification. Infill density variations can impact shape memory properties due to the shear strain induced by nozzle movement

Table 1. Default process parameters.

Parameters	Values
Layer height	0.12 mm
Raster angle	±45°
Print speed	50 mm/s
Printing temperature	200 °C
Nozzle size	0.4 mm
Retraction speed	25 mm/s
Initial layer thickness	0.3 mm

Table 2. Varied process parameters.

Input parameters	Different values considered
Infill density (%)	40, 70, and 100
Bed temperature (°C)	24, 30, 60, and 100

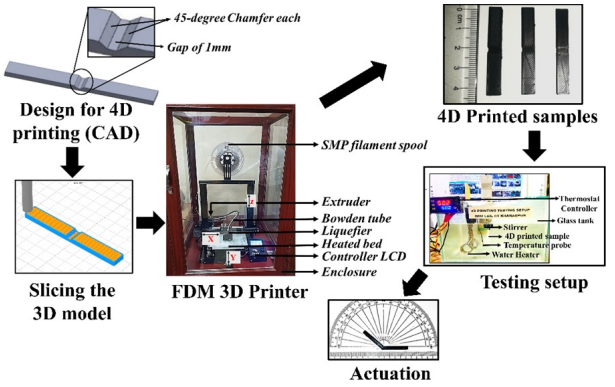


Fig. 1. Experimental methodology for shape memory property.

affecting material deposition direction, leading to deformation along the deposition path and thus affecting the infill density. Whereas the heated bed can alter pre-strain values by releasing thermal stresses, influencing shape recovery.

Thermo-mechanical programming in 4D printing is a process of training the shape memory material to remember its desired shape so that when an external fillip is applied, it can recover back to its trained state. Two different material programming conditions were applied to the samples as previously developed by us [26]. The first thermo-mechanical programming is called programming during printing. The PDP takes place during the processing of the SMP in FDM. As shown in Fig. 2(a), first, the SMP is fed through the extruder and heated to a temperature well above the T_g called programming temperature (T_p). Then, it gets fixed on the print bed, and further, due to the movement of the nozzle, the beads of the deposited polymer get stretched and induce pre-strain in the direction of the print. Finally, when the nozzle moves up by depositing multiple layers, the below layers get cooled (T_c), and thus, the programming of the SMP printed part is completed. Here, the programmed shape is the same as the FDM printed shape.

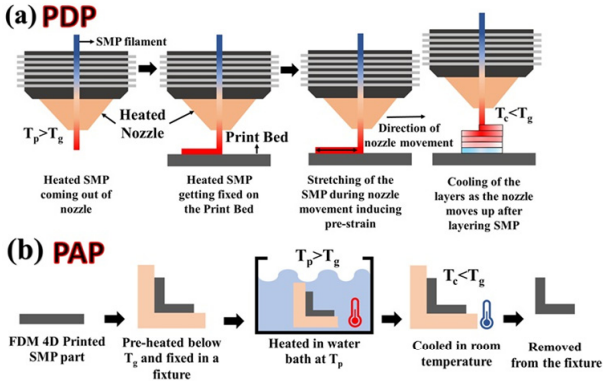


Fig. 2. Two different proposed thermo-mechanical programming for 4D printed parts.

The second thermo-mechanical programming is called programming after printing, as shown in Fig. 2(b). PAP has to be done separately after the printing as a post process. In PAP, the FDM 3D printed SMP is first preheated below T_g and mounted on a custom fixture that will program the sample at a 90-degree angle. The sample fixed in the fixture is then heated in hot water (in-house developed setup) above T_g called T_p of around 90 °C for 10 minutes. Then, the sample, along with the fixture, is removed from the hot water setup and kept at room temperature for 10 minutes to cool down (T_c). Finally, it is removed from the fixture, thereby getting the programmed part as 90 degrees.

Characterization of the shape memory properties is done as per the fold deployable test. Fold deployable angle measurement characterization is a robust, easy, and economical method for finding the shape fixity ratio (R_f), shape recovery ratio (R_r), and shape memory index (SMI). The process of angle measurement for both the programming methodologies is different and has been carried out as per our previous work [27], utilizing an in-house built testing setup [14]. For PDP, the samples are placed at a 90-degree angle for approximately 10 minutes in a fixture below their T_g (i.e. 50 °C). Subsequently, the samples are actuated in a testing setup at 80 °C, well above their T_g. Owing to residual strains from the printing process, the parts attempt to revert to their original printed position, recovering from the temporary deformation. Conversely, PAP is done after the FDM printing process. First, the samples are placed at a 90-degree angle in a fixture, and the shape is fixed below their T_g (i.e., 50 °C). Subsequently, the sample is exposed to a higher temperature (programming temperature (T_p) of 100 °C), which induces molecular alignment in the polymer, and then quenched to room temperature while maintaining constant applied stress. This completes the programming process, and the sample successfully retains its new programmed shape (90-degree) as its permanent form. Further, Eqs. (1)-(4) are used for calculating R_f, R_r, SMI.

$$R_f = \frac{\theta_{fix}}{\theta_{max}} \quad (1)$$

$$R_r = \frac{\theta_{fix} - (\theta_{end} + (90 - \theta_{def}))}{\theta_{fix}} \quad (2)$$

$$R_r = \frac{\theta_{fix} - (\theta_{end} + (180 - \theta_{def}))}{\theta_{fix}} \quad (3)$$

$$SMI = R_r \times R_f \quad (4)$$

Where, R_r is the recovery ratio, R_f is the fixity ratio, θ_{max} is the programmed angle, θ_{fix} is the fixed angle, θ_{end} is the recovered angle, and θ_{def} is the deformation angle.

3. Results and discussion

The study demonstrates that shape recovery varies across different values of infill density and bed temperature, as illustrated in Table 3. Table 3(a) depicts the shape recovery of 4D printed samples for the PDP method, while Table 3(b) displays the shape recovery in samples programmed using the PAP method. Further, it is found that there is a direct relationship between infill density and shape recovery. Our results consistently demonstrated that as infill density increased within the printed parts, shape recovery also improved (Figs. 3(a) and (d)). This phenomenon can be attributed to the presence of gaps between consecutive material layers in parts with lower infill. These gaps, arising from reduced infill and material content, led to a corresponding reduction in the accumulation of residual stresses within the components.

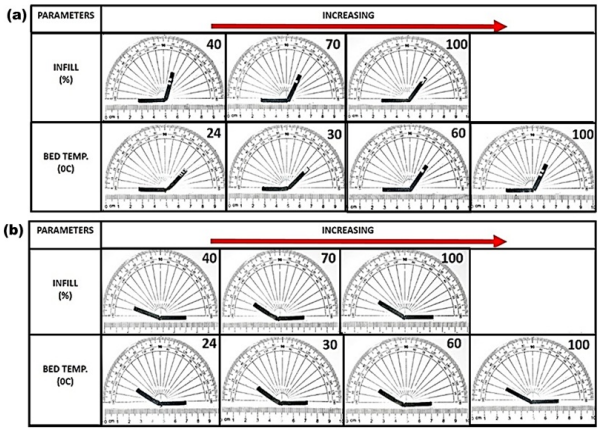
Another significant outcome of our study pertains to the influence of bed temperature on shape recovery. It was consistently observed that elevating the bed temperature resulted in a decrease in shape recovery (Figs. 3(b) and (c)). This is primarily due to the bed temperature's effect on slowing down the cooling process, which, in turn, reduces the level of residual strain within the printed components. Furthermore, the higher bed temperature acted as a mitigating factor for certain thermal stresses, ultimately leading to a decreased extent of shape recovery in the printed parts. The result obtained is consistent with the results previously obtained by other researchers [23].

The shape fixity ratio demonstrates consistent performance, with minimal deviations across various process parameters. There is a subtle decline in the shape fixity ratio at higher infill, whereas, it increases with elevated bed temperatures. It is worth noting that these changes are relatively minor and may be attributed to the relatively small thickness of the parts examined in our experiments. Further investigation involving thicker components is warranted to gain deeper insights into the influence of process parameters on the shape fixity ratio.

The analysis between the two different programming methods by considering the SMI, which takes into account both the R_f and R_r, exhibits a significantly higher value in the samples that were programmed during the printing process (PDP) compared to those re-programmed after printing (PAP). This disparity can likely be attributed to the re-programming procedure, which involves heating the sample much above the T_g to relieve some of the residual strain accumulated during the initial printing. Although subsequent quenching at room temperature induces a degree of strain, this is insufficient as compared to the pre-strain generated during the FDM process. As a result, the average SMI percentage for PDP samples surpasses that of PAP samples by approximately 40 % for both the process parameters (Tables 4 and 5).

This research explores the relationship between infill density,

Table 3. Samples after recovery for various process parameters considered: (a) PDP; (b) PAP.



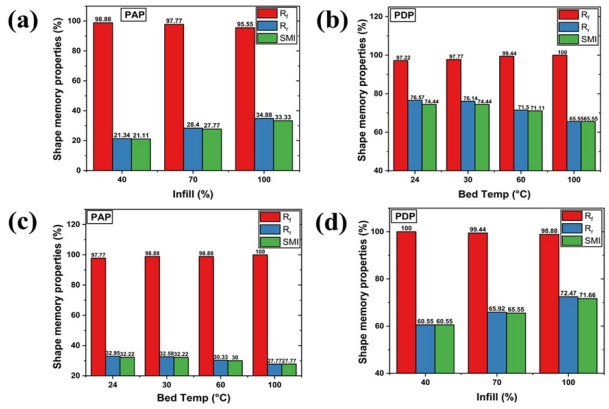


Fig. 3. Variation in shape memory properties for various printing process parameters and across multiple programming conditions.

Table 4. Shape memory properties for PDP.

Infill (%)	θ_{max}	θ_{fix}	θ_{end}	R_f (%)	R_r (%)	SMI (%)
40	180	180	71	100	60.55	60.55
70	180	179	61	99.44	65.92	65.55
100	180	178	49	98.88	72.47	71.66
Bed temp. (°C)	θ_{max}	θ_{fix}	θ_{end}	R_f (%)	R_r (%)	SMI (%)
24	180	175	41	97.22	76.57	74.44
30	180	176	42	97.77	76.14	74.44
60	180	179	51	99.44	71.50	71.11
100	180	180	62	100	65.55	65.55

Table 5. Shape memory properties for PAP.

Infill (%)	θ_{max}	θ_{fix}	θ_{end}	R_f (%)	R_r (%)	SMI (%)
40	90	89	70	98.88	21.34	21.11
70	90	88	63	97.77	28.40	27.77
100	90	86	56	95.55	34.88	33.33
Bed temp. (°C)	θ_{max}	θ_{fix}	θ_{end}	R_f (%)	R_r (%)	SMI (%)
24	90	88	59	97.77	32.95	32.22
30	90	89	60	98.88	32.58	32.22
60	90	89	62	98.88	30.33	30
100	90	90	65	100	27.77	27.77

bed temperature, and shape memory properties in FDM 4D printing, which can have significant applications in various industries. The ability to control shape morphing using thermo-mechanical programming (PAP & PDP) with printing process parameters can lead to highly customizable and functional dynamic components. This can be applied in aerospace engineering to create morphing wings that can adapt to external stimuli, enhancing fuel efficiency and performance. In healthcare, 4D printing can create patient-specific medical implants with tailored shape memory, ensuring minimal invasive deployment, optimal fit and function. In robotics, FDM 4D printed parts can be integrated to create soft, adaptive robots capable of executing complex tasks in changing environments with exceptional efficiency.

4. Conclusions

This study has revealed the complex relationship between infill density, bed temperature, and programming methodology in shape memory properties of 4D-printed PLA-SMP. Higher infill densities have a positive effect, while elevated bed temperatures have a negative impact. The highest recovery ratio of around 76 % is achieved for a bed temperature of 24 °C, i.e., no external heating of the bed is done. Similarly, the highest recovery ratio for the 100 % infill density is around 72 %. These high shape recovery values are obtained for the PDP thermo-mechanical programming method. In comparison, between the two programming methods, PDP yields 40 % more shape recovery than PAP. This highlights the importance of precise parameter control as well as the influence of various programming processes in achieving a high shape recovery ratio and controlled shape morphing behavior. This experimental research study establishes the groundwork for enhancing and tailoring the shape memory properties in 4D-printed SMPs fabricated via FDM.

As the study is limited by focusing on PLA-based SMP material and FDM printing, with only two process parameters and thermo-mechanical programming methods applied. Future research could expand to include a broader range of smart materials and 3D printing processes. Exploring alternative programming methods and optimizing various process parameters can improve shape memory properties. Moreover, validating the findings in real-world applications in the areas of aerospace, healthcare, and robotics would enhance the understanding and future potential of 4D printing technology.

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References

[1] S. Jayachandran et al., Investigations on performance viability of NiTi, NiTiCu, CuAlNi and CuAlNiMn shape memory alloy/kapton composite thin film for actuator application, *Composites Part B: Engineering*, 176 (2019) 107182, doi: 10.1016/j.compositesb.2019.107182.

[2] W. Sokolowski, A. Metcalfe, S. Hayashi, L. Yahia and J. Raymond, Medical applications of shape memory polymers, *Biomedical Materials*, 2 (1) (2007) S23, doi: 10.1088/1748-6041/2/1/S04.

[3] Y. Xia, Y. He, F. Zhang, Y. Liu and J. Leng, A review of shape memory polymers and composites: mechanisms, materials, and applications, *Advanced Materials*, 33 (6) (2021) 2000713, doi: https://doi.org/10.1002/adma.202000713.

[4] A. S. Gladman, E. A. Matsumoto, R. G. Nuzzo, L. Mahadevan and J. A. Lewis, Biomimetic 4D printing, *Nature Materials*, 15 (4) (2016) 413-418, doi: 10.1038/NMAT4544.

[5] N. A. Meisel, A. M. Elliott and C. B. Williams, A procedure for creating actuated joints via embedding shape memory alloys in PolyJet 3D printing, *Journal of Intelligent Material Systems and Structures*, 26 (12) (2015) 1498-1512, doi: 10.1177/1045389X14544144.

[6] L. Yue et al., Cold-programmed shape-morphing structures based on grayscale digital light processing 4D printing, *Nature Communications*, 4 (1) (2023) 5519, doi: 10.1038/s41467-023-41170-4.

[7] P. Wang et al., Enhanced multi-material 4D printing hybrid composites based on shape memory polymer/thermoplastic elastomer, *Smart Materials and Structures*, 32 (5) (2023) 55025, doi: 10.1088/1361-665X/acc914.

[8] Z. Lyu, J. Wang and Y. Chen, 4D printing: interdisciplinary integration of smart materials, structural design, and new functionality, *International Journal of Extreme Manufacturing*, 5 (3) (2023) 32011, doi: 10.1088/2631-7990/ace090.

[9] A. Cano-Vicent et al., Fused deposition modelling: Current status, methodology, applications and future prospects, *Additive Manufacturing*, 47 (2021) 102378, doi: https://doi.org/10.1016/j.addma.2021.102378.

[10] W. M. Huang, Z. Ding, C. C. Wang, J. Wei, Y. Zhao and H. Purnawali, Shape memory materials, *Materials Today*, 13 (7) (2010) 54-61, doi: https://doi.org/10.1016/S1369-7021(10)70128-0.

[11] S. Tibbits, *The Emergence of "4D Printing"*, TED (2013) Available at: https://www.ted.com/talks/skylar_tibbits_the_emer

gence _of_4d_printing.
[12] S. Tibbits, 4D printing: Multi-material shape change, *Architectural Design*, 84 (1) (2014) 116-121, doi: 10.1002/AD.1710.
[13] M. Bodaghi, A. R. Damanpack and W. H. Liao, Adaptive metamaterials by functionally graded 4D printing, *Materials and Design*, 135 (2017) 26-36, doi: 10.1016/j.matdes.2017.08.069.
[14] B. B. Samal, A. Jena, S. K. Varshney and C. S. Kumar, 4D printing: An experimental case study on processing of shape memory polymer by FDM/FFF for nature inspired structures, A. Kumar, R. K. Mittal, and A. Haleem (Eds.), *Advances in Additive Manufacturing Artificial Intelligence, Nature-Inspired, and Biomanufacturing*, Elsevier (2023) 361-377, doi: 10.1016/B978-0-323-91834-3.00019-3.
[15] R. Noroozi, M. Bodaghi, H. Jafari, A. Zolfagharian and M. Fotouhi, Shape-adaptive metastructures with variable bandgap regions by 4D printing, *Polymers*, 12 (3) (2020) 519, doi: 10.3390/polym12030519.
[16] W. Wu, W. Ye, Z. Wu, P. Geng, Y. Wang and J. Zhao, Influence of layer thickness, raster angle, deformation temperature and recovery temperature on the shape-memory effect of 3D-printed polylactic acid samples, *Materials*, 10 (8) (2017) 970, doi: 10.3390/ma10080970.
[17] M. Hosseinzadeh, M. Ghoreishi and K. Narooei, An investigation into the effect of thermal variables on the 3D printed shape memory polymer structures with different geometries, *Journal of Intelligent Material Systems and Structures*, 33 (5) (2022) 715-726, doi: 10.1177/1045389X211028286.
[18] A. R. Rajkumar and K. Shanmugam, Additive manufacturing-enabled shape transformations via FFF 4D printing, *Journal of Materials Research*, 33 (24) (2018) 4362-4376, doi: 10.1557/jmr.2018.397.
[19] M. Barletta, A. Gisario and M. Mehrpouya, 4D printing of shape memory polylactic acid (PLA) components: Investigating the role of the operational parameters in fused deposition modelling (FDM), *Journal of Manufacturing Processes*, 61 (2021) 473-480, doi: https://doi.org/10.1016/j.jmapro.2020.11.036.
[20] P. Wu, T. Yu, M. Chen and D. Hui, Effect of printing speed and part geometry on the self-deformation behaviors of 4D printed shape memory PLA using FDM, *Journal of Manufacturing Processes*, 84 (2022) 1507-1518, doi: https://doi.org/10.1016/j.jmapro.2022.11.007.
[21] M. Hosseinzadeh, M. Ghoreishi and K. Narooei, 4D printing of shape memory polylactic acid beams: an experimental investigation into FDM additive manufacturing process parameters, mathematical modeling, and optimization, *Journal of Manufacturing Processes*, 85 (2023) 774-782, doi: https://doi.org/10.1016/j.jmapro.2022.12.006.
[22] Y. S. Alshebly and M. Nafea, Effects of printing parameters on 4D-printed PLA actuators, *Smart Materials and Structures*, 32 (6) (2023) 064008, doi: 10.1088/1361-665X/acd504.
[23] S. M. D. Tezerjani, M. S. Yazdi and M. H. Hosseinzadeh, The effect of 3D printing parameters on the shape memory properties of 4D printed polylactic acid circular disks: an experimental investigation and parameters optimization, *Materials Today Communications*, 33 (2022) 104262, doi: https://doi.org/10.1016/j.mtcomm.2022.104262.
[24] S. Nam and E. Pei, The influence of shape changing behaviors from 4D printing through material extrusion print patterns and infill densities, *Materials*, 13 (17) (2020) 3754, doi: 10.3390/ma13173754.
[25] eSUN, *PLA+*, eSUN, https://www. esun3d.com/pla-pro-product/.
[26] B. B. Samal, A. Jena, S. K. Varshney and C. S. Kumar, FDM 4D printing: a low-cost approach of shape programming and assessing the shape memory properties using angle measurement methods in hot water actuation testing apparatus, *Materials Today: Proceedings* (2023) doi: 10.1016/j.matpr.2023.02.038.
[27] B. B. Samal, A. Jena, S. K. Varshney and C. S. Kumar, 4D printing of shape memory polymers: a comparative study of programming methodologies on various material properties, *Smart Materials and Structures*, 32 (7) (2023) 074003, doi: 10.1088/1361-665X/acda6e.



Bijaya Bikram Samal is a Ph.D. Research Scholar at the Advanced Technology Development Centre, Indian Institute of Technology, Kharagpur, West Bengal, India. He obtained M.Tech Gold Medal in Production Engineering from Veer Surendra Sai University of Technology, Odisha, India, and a B.Tech in Mechanical Engineering from Biju Patnaik University of Technology, Odisha, India. He is recognized as the first person in India to have initiated research in the area of 4D printing. He is a Chartered Engineer (Mechanical) of Institute of Engineers. He is a lead author of the book "4D Printing Technology: Principles, Materials and Application". His research interests include 4D printing, additive manufacturing, laser material processing, and micro-manufacturing.



Shailendra Kumar Varshney is a Professor of the Department of Electronics and Electrical Communication Engineering, Indian Institute of Technology, Kharagpur, West Bengal, India. He obtained his Ph.D. in Applied Physics from the University of Delhi in 2005. He obtained M.Sc. Gold Medal in Physics from Aligarh Muslim University in 1999. He has received numerous prestigious fellowships, including the UGC/CSIR JRF fellowship from India, the Monbukagakusho and JSPS fellowship from Japan, and the Alexander von Humboldt fellowship from Germany. His research focuses on PCFs, on-chip photonic components, nonlinear photonics, and quantum photonics.



Cheruvu Siva Kumar is a Professor of the Mechanical Engineering Department at Indian Institute of Technology, Kharagpur, West Bengal, India. He has obtained his B.Tech. in Manufacturing Science and Engineering in 1987 and a Ph.D. in Mechanical Engineering in 1995, both from IIT Kharagpur. He conducted post-doctoral research (JSPS Fellow) at AIST, Japan. He specializes in Additive Manufacturing, Robotics, CAD-CAM, and Computer Networks. Engaged in numerous national projects, he has over 100 publications and 10 patents. Additionally, he serves as an expert panelist in governmental projects.